COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures And Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	08/26/26	**DATE SUBMITTED:**	08/26/26
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1. PROCEDURE OUTPUT

Array-Based Stack:

Algorithm:
Since its LIFO ill just pop it out then save it onto a temp variable then compare the first stack then check the temp stack if its equal to the first stack.

First_stack == temp_stack

Return true;

Queue Using LinkedList:

```
template <typename T>
void checker(qNode<T> **front, qNode<T> **back)
{
    if ((*front)->data == (*back)->data)
    {
        std::cout << "Symmetrical" << std::endl;
    }

    else
    {
        std::cout << "Not Symmetrical" << std::endl;
    }
}
```

I came up with first it checks the front data if its equal to the back data if its equal then its symmetrical. Its kinda bruteforced since if if its like 12351 it would say its

symmetrical even thought it is not because in this function it only checks the front and back.

Sample Test Data

Input Sequence	Symmetric?	Output (Console)	Remark
1 2 3 2 1	Yes		Partially Success
1 2 3 4 5	No		Partially Success
1 2 2 1	Yes		Partially Success
7	Yes		Partially Success
1 2 3 3 2 1	Yes		Partially Success
1 2 3 4 3 2 5	No		Partially Success

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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